软件学院综合性、设计性实验报告

专业:计算机科学与技术 班级:18 计科移动开发班 2020—2021 学年第一学期

课程名称	编译原理	指导教师	张聪品
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实验地点	过街楼 404 实验室	实验时间	2020.10.01
项目名称	词法分析器的构造	实验类型	综合性/设计性

一、实验目的

设计并实现一个词法分析器，实现对源程序文本文件的读取，并能够对该源程序中的所有单词进行分类，指出其所属类型，实现简单的词法分析操作。

二、实验仪器或设备

实验环境:Microsoft Visual Studio 2012

编程环境:C++

三、总体设计（设计原理、设计方案及流程等）

待检测的源语言为类 C 语言(即缺少函数、语法识别的 C 代码)

（1）能够将源语言输出

（2）能够将该源程序中所有的单词根据其所属类型（整数、保留字、运算符、标识符等。定义的类 C 语言中的标识符只能以字母或下划线开头）进行归类显示，例如：识别保留字：if、int、for、while、do、return、break、continue 等，其他的都识别为标识符；常数为无符号整形数；运算符包括：+、-、*、/、=、>、<、>=、<=、!=等；分隔符包括：,、;、{、}、(、)等。

四、实验步骤（包括主要步骤、代码分析等）

```
//main.cpp
#include<iostream>
#pragma warning (disable:4996)
using namespace std;
#define MAX 10
/*
保留字|关键字：1
操作符|运算符：2
分界符：3
标识符：4
常数：5
无识别：6
*/
char ch = ' ';
char*keyWord[10]={ "void","main","break","include","begin","end","if","else","while","switch" };
char token[20];//定义获取的字符
                //判断是否是关键字
```

```
bool isKey(char * token)
{
    for (int i = 0; i < MAX; i++)
    {
        if (strcmp(token, keyWord[i]) == 0)
            return true;
    }
    return false;
}
//判断是否是字母
bool isLetter(char letter)
{
    if ((letter >= 'a' && letter <= 'z') || (letter >= 'A' && letter <= 'Z'))
        return true;
    else
        return false;
}
//判断是否是数字
bool isDigit(char digit)
{
    if (digit >= '0' && digit <= '9')
        return true;
    else
        return false;
}
//词法分析
void analyze(FILE *fpin)
{
    while ((ch = fgetc(fpin)) != EOF) {
        if (ch == ' ' || ch == '\t' || ch == '\n') {}
        else if (isLetter(ch)) {
            char token[20] = { '\0' };
            int i = 0;
            while (isLetter(ch) || isDigit(ch)) {
                token[i] = ch;
                i++;
                ch = fgetc(fpin);
            }
            //回退一个指针
            fseek(fpin, -1, SEEK_CUR);
            if (isKey(token)) {
                //关键字
                cout << token << "\t1" << "\t 关键字" << endl;
            }
```

```
        else {
            //标识符
            cout << token << "\t4" << "\t 标识符" << endl;
        }
    }
    else if (isDigit(ch) || (ch == '.'))
    {
        int i = 0;
        char token[20] = { '\0' };
        while (isDigit(ch) || (ch == '.'&&isDigit(fgetc(fpin))))
        {
            if (ch == '.')fseek(fpin, -1, SEEK_CUR);
            token[i] = ch;
            i++;
            ch = fgetc(fpin);
        }
        fseek(fpin, -1, SEEK_CUR);
        //属于无符号常数
        cout << token << "\t5" << "\t 常数" << endl;
    }
    else switch (ch) {
        //运算符
    case '+': {
        ch = fgetc(fpin);
        if (ch == '+')cout << "++" << "\t2" << "\t 运算符" << endl;
        if (ch == '=')cout << "+=" << "\t2" << "\t 运算符" << endl;
        else {
            cout << "+" << "\t2" << "\t 运算符" << endl;
            fseek(fpin, -1, SEEK_CUR);
        }
    }break;
    case '-': {
        ch = fgetc(fpin);
        if (ch == '-')cout << "--" << "\t2" << "\t 运算符" << endl;
        if (ch == '=')cout << "-=" << "\t2" << "\t 运算符" << endl;
        else {
            cout << "-" << "\t2" << "\t 运算符" << endl;
            fseek(fpin, -1, SEEK_CUR);
        }
    }break;

    case '*': {
        ch = fgetc(fpin);
        if (ch == '=')cout << "*=" << "\t2" << "\t 运算符" << endl;
```

```
        else
        {
            cout << ch << "\t2" << "\t 运算符" << endl;
            fseek(fpin, -1, SEEK_CUR);
        }
    } break;

    case '/': {
        ch = fgetc(fpin);
        if (ch == '=')cout << "/=" << "\t2" << "\t 运算符" << endl;
        else
        {
            cout << ch << "\t2" << "\t 运算符" << endl;
            fseek(fpin, -1, SEEK_CUR);
        }
    } break;
        //分界符
    case '(':cout << ch << "\t3" << "\t 分界符" << endl; break;
    case ')':cout << ch << "\t3" << "\t 分界符" << endl; break;
    case '[':cout << ch << "\t3" << "\t 分界符" << endl; break;
    case ']':cout << ch << "\t3" << "\t 分界符" << endl; break;
    //case ';':cout << ch << "\t3" << "\t 分界符" << endl; break;
    case '{':cout << ch << "\t3" << "\t 分界符" << endl; break;
    case '}':cout << ch << "\t3" << "\t 分界符" << endl; break;
        //运算符
    case '=': {
        ch = fgetc(fpin);
        if (ch == '=')cout << "==" << "\t2" << "\t 运算符" << endl;
        else {
            cout << "=" << "\t2" << "\t 运算符" << endl;
            fseek(fpin, -1, SEEK_CUR);
        }
    }break;
    case ':': {
        ch = fgetc(fpin);
        if (ch == '=')cout << ":=" << "\t2" << "\t 运算符" << endl;
        else {
            cout << ":" << "\t2" << "\t 运算符" << endl;
            fseek(fpin, -1, SEEK_CUR);
        }
    }break;
    case '>': {
        ch = fgetc(fpin);
        if (ch == '=')cout << ">=" << "\t2" << "\t 运算符" << endl;
```

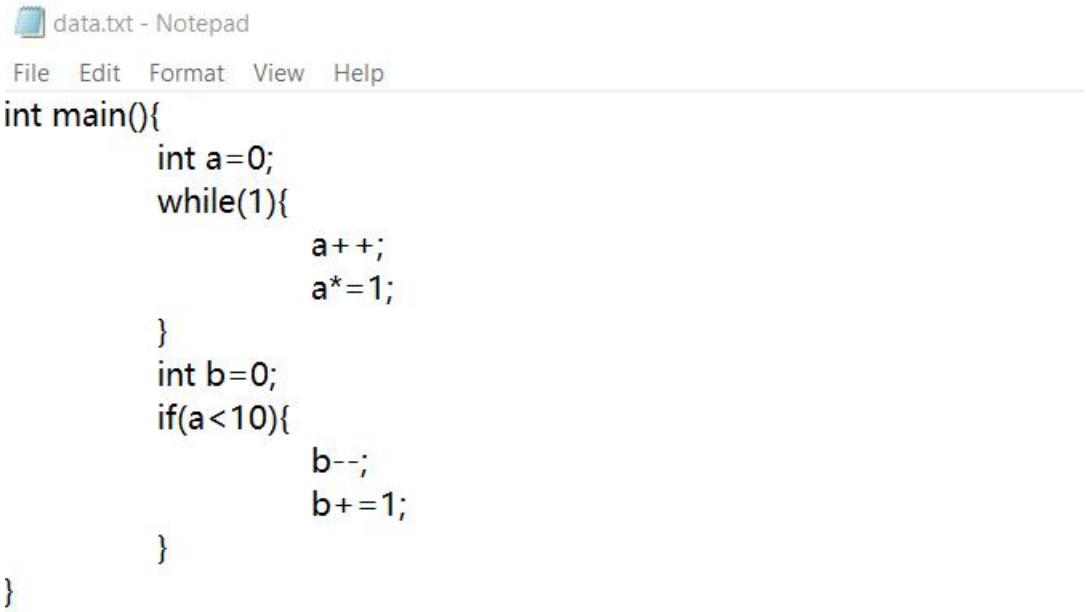
```
            else {
                cout << ">" << "\t2" << "\t 运算符" << endl;
                fseek(fpin, -1, SEEK_CUR);
            }
        }break;
        case '<': {
            ch = fgetc(fpin);
            if (ch == '=')cout << "<=" << "\t2" << "\t 运算符" << endl;
            else {
                cout << "<" << "\t2" << "\t 运算符" << endl;
                fseek(fpin, -1, SEEK_CUR);
            }
        }break;
            //无识别
        default: cout << ch << "\t6" << "\t 无识别符" << endl;
        }
    }
}
int main() {
    char input[30];
    FILE *fpin;
    cout << "请输入源文件名：\n" << endl;
    for (;;) {
        cin >> input;
        if ((fpin = fopen(input, "r")) != NULL)
            break;
        else
            cout << "路径输入错误" << endl;
    }
    cout << "**********词法分析结果**********" << endl;
    analyze(fpin);
    fclose(fpin);
}
//wrUtils.h

#pragma once
#include<iostream>
#include<fstream>
#include<string>
#include<cassert>
#include<vector>
#include<stdio.h>
using namespace std;
```

```
char split(char *a, char b) {
	int tag;
	tag=strlen(a)+1;
	char* buff = new char[tag];
	for (int i = 0; i < strlen(a); i++) {
		buff[i] = a[i];
	}
	for (int i = 0; i < 1; i++) {
		buff[i + strlen(a)] = b;
	}
	return *buff;
}

vector<string>getKWList(){
	vector<string>keyWordList;
	//keyWordList = { "False", "None", "True", "and", "as", "assert", "break",
"class", "continue", "def", "del", "elif", "else", "except", "finally", "for", "from",
"global", "if", "import", "in", "is", "lambda", "nonlocal", "not", "or", "pass", "raise",
"return", "try", "while", "with", "yield" };
	return keyWordList;
}

char *getSymbolList() {
	char symbolList[] =
{ 32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,58,59,60,61,62,63,64,91,92,93,94,95,96,
123,124,125,126 };
	return symbolList;
}


int getWordCount(string file) {
	ifstream ifs;
	char ch;
	ifs.open(file);
	int tag = 0;
	while (ifs.get(ch)) {
		tag++;
	}
	ifs.close();
	return tag;
}
```

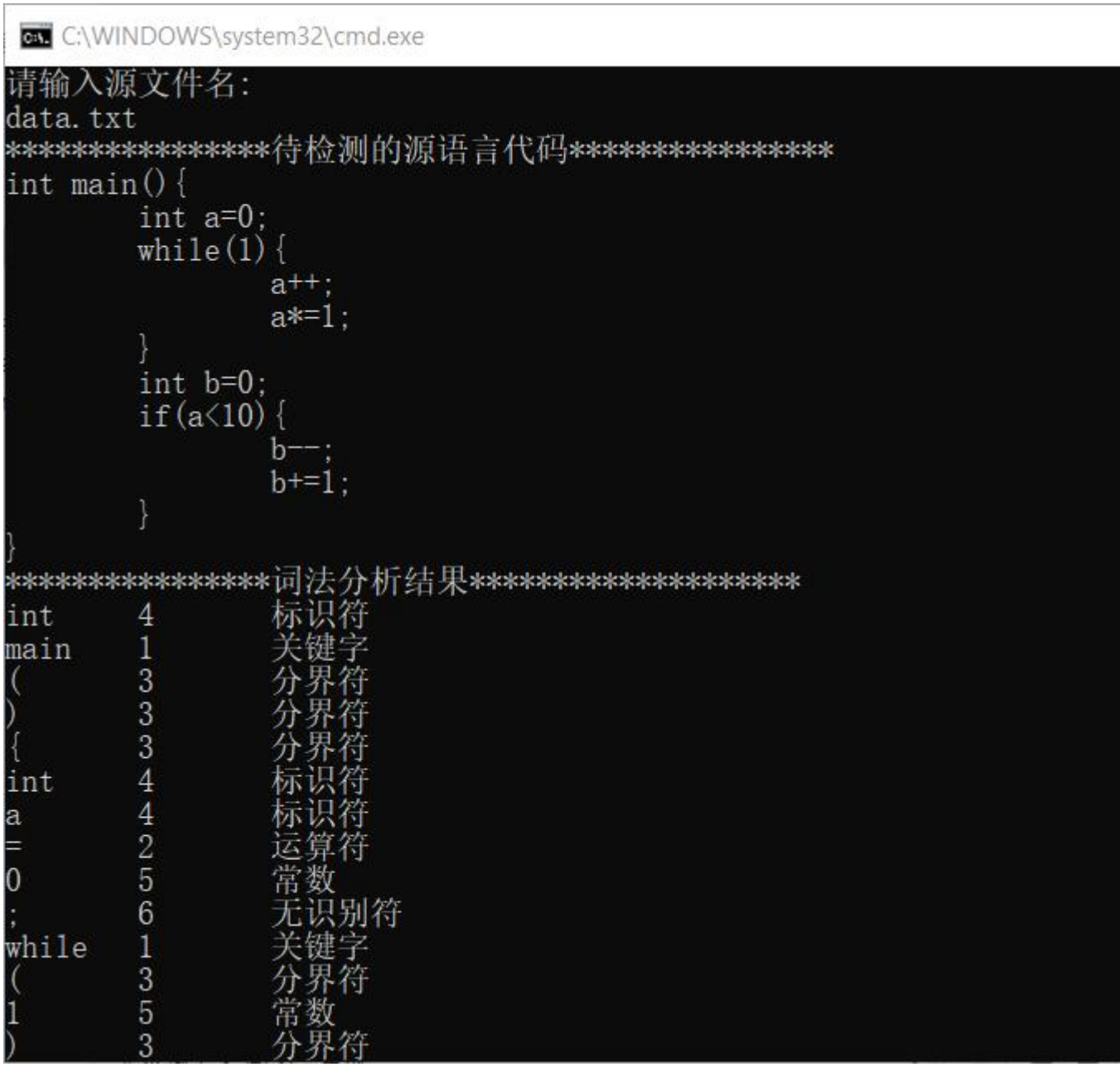
```
vector<string>getPyKwlist() {
	string filename = "pyKWlist.txt";
	vector<string>keyWordList;
	int tag = getWordCount(filename);
	char *p,ch;
	p = new char[tag];
	ifstream file;
	file.open(filename);
	int a = 0;
	while (file.get(ch) && a < tag) {
		p[a]= ch;
		a++;
	}/*
	char reg = ',';
	char *all = new char[tag];
	for (int i = 0; i < tag; i++) {
		if (p[i]!=reg) {
			*all=split(all,p[i]);
		}
	}
	for (int i = 0; i < strlen(all);i++){
		cout << all[i];
	}
	*/
	//keyWordList = { "False", "None", "True", "and", "as", "assert", "break",
"class", "continue", "def", "del", "elif", "else", "except", "finally", "for", "from",
"global", "if", "import", "in", "is", "lambda", "nonlocal", "not", "or", "pass", "raise",
"return", "try", "while", "with", "yield" };
	return keyWordList;
}

void write(string file, string data) {
	ofstream ofs;
	ofs.open(file, ios::out);
	ofs << data << endl;
}

void readLine(string file) {
	ifstream infile;
	infile.open(file.data());	//将文件流对象与文件连接起来
	assert(infile.is_open());	//若失败,则输出错误消息,并终止程序运行

	string s;
	while (getline(infile, s))
```

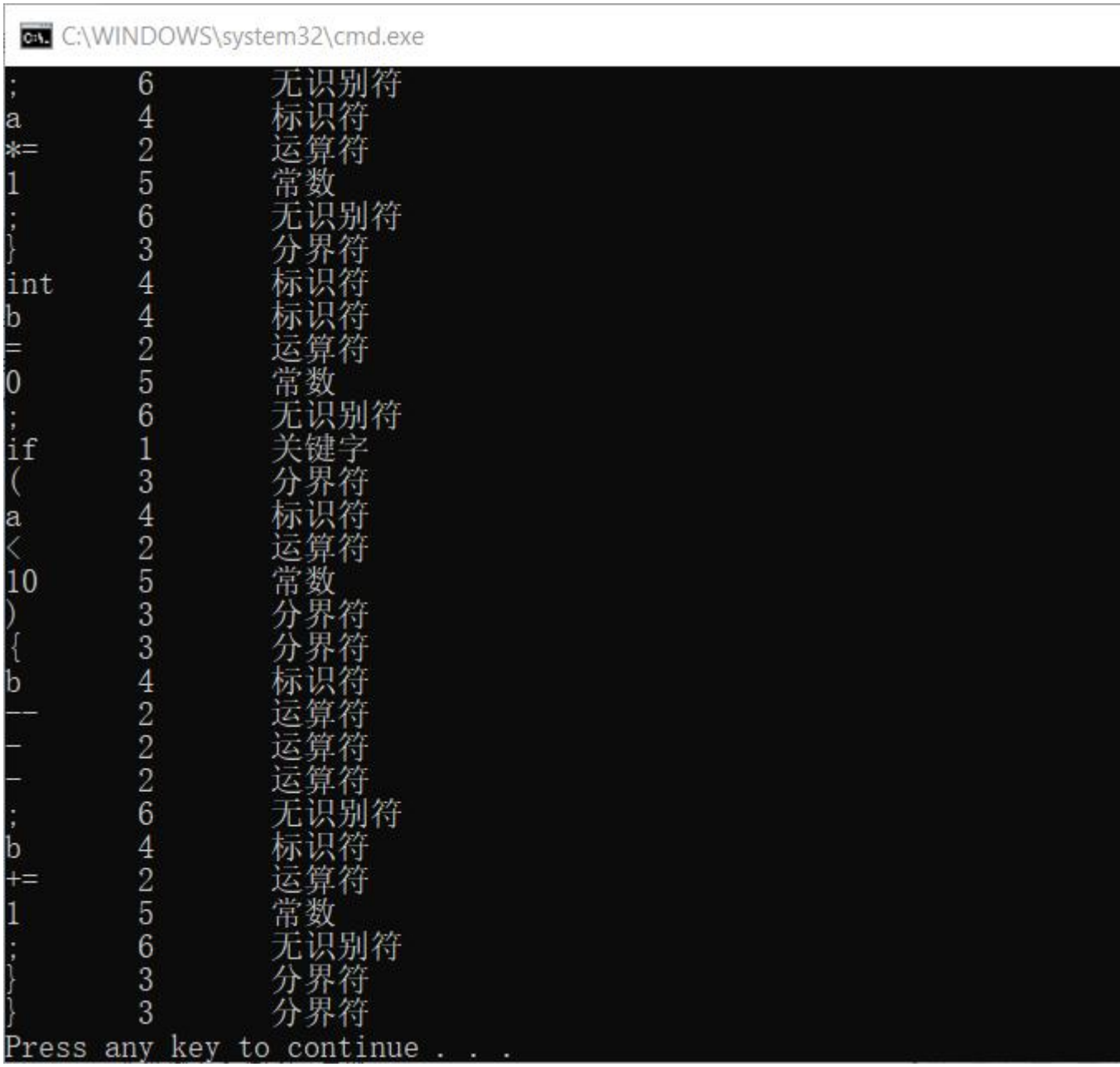
```
    {
        cout << s << endl;
    }
    infile.close();                    //关闭文件输入流
}

void readWithSkip(string file)
{
    ifstream infile;
    infile.open(file.data());     //将文件流对象与文件连接起来
    assert(infile.is_open());     //若失败,则输出错误消息,并终止程序运行

    char c;
    infile >> noskipws;
    while (!infile.eof()){
        infile >> c;
        cout << c << endl;

    }
    infile.close();                    //关闭文件输入流
}

void readWithoutSkip(string file)
{
    ifstream infile;
    infile.open(file.data());     //将文件流对象与文件连接起来
    assert(infile.is_open());     //若失败,则输出错误消息,并终止程序运行

    char c;
    while (!infile.eof())
    {
        infile >> c;
        cout << c << endl;

    }
    infile.close();                    //关闭文件输入流
}


//待识别的源语言代码
//data.c
int main(){
    int a=0;
    while(1){
```

```
        a++;
        a*=1;
    }
    int b=0;
    if(a<10){
        b--;
        b+=1;
    }
}
```



五、实验结果


C:\WINDOWS\system32\cmd.exe
请输入源文件名:
data.txt
**********待检测的源语言代码**********
int main(){
int a=0;
while(1){
a++;
a*=1;
}
int b=0;
if(a<10){
b--;
b+=1;
}
}
**********词法分析结果**********
int 4 标识符
main 1 关键字
(3 分界符
) 3 分界符
{ 3 分界符
int 4 标识符
a 4 标识符
= 2 运算符
0 5 常数
; 6 无识别符
while 1 关键字
(3 分界符
1 5 常数
) 3 分界符


C:\WINDOWS\system32\cmd.exe
; 6 无识别符
a 4 标识符
*= 2 运算符
1 5 常数
; 6 无识别符
} 3 分界符
int 4 标识符
b 4 标识符
= 2 运算符
0 5 常数
; 6 无识别符
if 1 关键字
(3 分界符
a 4 标识符
< 2 运算符
10 5 常数
) 3 分界符
{ 3 分界符
b 4 标识符
-- 2 运算符
- 2 运算符
- 2 运算符
; 6 无识别符
b 4 标识符
+= 2 运算符
1 5 常数
; 6 无识别符
} 3 分界符
} 3 分界符
Press any key to continue . . .

六、总结

通过这次实验清晰的理解了词法分析器的原理，对现代编译原理技术有了清晰的了解，为学习语法分析打好了基础

教师签名:

年　　月　　日